
Exploring ViT Architectures for Enhanced Classification

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Abstract

This paper explores different Vision Transformer (ViT) architectures for image classification on the CIFAR-100 dataset, comparing four models: a baseline ViT, a pretrained ViT masked Autoencoder (MAE) + fine-tuning, a joint ViT with random masking, and a joint ViT with importance-based masking. I evaluate these models on classification accuracy and reconstruction loss, training each for 50 epochs. The results show that the joint ViT with random masking achieved the highest test accuracy at 42.16%, while the importance joint ViT model performed a tick below at 41.98%. The challenges are that transformer models often require large-scale datasets, longer training and careful hyperparameter tuning compared to Convolutional Neural Networks (CNN), which hinders the models' ability to generalize especially without a considerable amount of pretraining. This limitation likely contributes to the low test accuracy on CIFAR-100. This paper demonstrates the potential of joint training in ViTs and introduces an importance based masking approach which will lay the groundwork for future innovations of ViT architectures on small datasets.

Code: [Here](#)

Introduction

Vision Transformers have revolutionized image processing since their introduction in 2020, adapting the transformer architecture, which was originally used for NLP tasks to process images. Unlike convolutional neural networks (CNNs), which rely on applying filters on local patches of an image and capture spatial information in a hierarchical manner, ViTs divide images into patches, embed them into tokens, and use a mechanism which allows each patch to attend to every other patch, allowing global context to be captured from the first layer. This makes ViTs effective for image classification tasks, but their performance often relies on large-scale pre-training, which may result in overfitting when labeled data is scarce.

The rise of self-supervised learning (SSL) has been an

effective approach to tackle this challenge because this idea enables models to learn meaningful representations from unlabeled data, reducing the reliance of labeled datasets. Ideas such as Masked Autoencoders (MAE), introduced by He et al. in 2022, have become pivotal for ViTs, where the model is trained to reconstruct random image patches, which creates an understanding of the image context [1]. MAE pre-training presents evidence that ViTs can achieve results in tasks such as classification after fine-tuning [1].

Joint mask reconstruction and classification presents a very creative multitasking approach for ViTs, combining self-supervised reconstruction tasks with a supervised classification task. This method trains the ViT encoder to categorize images and restore masked sections of those images simultaneously, therefore allowing the extraction of better features and inherent biases from the data without requiring additional labeled data. The ViT simultaneously can predict class labels while reconstructing image regions, which can be derived from MAE.

This paper explores different strategies to optimize ViT architectures for small datasets, exclusively focusing on the CIFAR-100 dataset. The scope of this project follows the design, implementation, and evaluation of four different ViT models: A baseline ViT trained from scratch, a pretrained ViT with fine tuning, a joint ViT with random masking, and a joint ViT with importance-based masking. The key objectives are to compare the different classification accuracies, reduce reliance on large, labeled datasets, and to address the high computational cost of training ViTs as all training is done on a NVIDIA Tesla T4 GPU. The research in this project is significant as it currently aligns with research efforts towards scalable and efficient transformer models as it serves as an alternative to CNNs for real world applications requiring efficient vision systems. The approach outlined in this paper combines established techniques like MAE with an innovative masking strategy which I call importance masking, offering a comparative analysis which will guide future optimizations in ViT architectures.

Related Work

The development of ViTs by Dosovitskiy in 2020 has shifted the landscape of image classification, primarily offering an alternative to CNNs. This project builds on the recent advancements in self-supervised learning and efficiency opti-

mization, drawing inspiration from three different research papers.

Masked autoencoders for Self-Supervised Learning

He et al [1] introduced the concept of Masked Autoencoders (MAE), which is defined as a simple but very powerful self-supervised training scheme for Vision Transformers (ViT). MAEs essentially mask a large random portion of the patches of an input image (e.g. 75%) and fill in the missing pixels by training a model which predicts the appearance of those missing pixels. This design choice ultimately enables a more efficient training of large models, since the encoder only sees 25% of patches during each iteration. Pre-training is approximately 3x faster and uses less memory, which allows the model to reach higher capacity, using less compute power. After self-supervised pre-training on ImageNet-1K, the decoder is not used, and the encoder is fine-tuned in the output of prior tasks. MAEs can enable ViTs to generalize relationships using unlabeled data and high-capacity models, achieving great accuracy without additional labeled data.

This paper directly inspired my MAE + fine tuning model where I adopt the MAE pre-training to do feature learning on the CIFAR-100 dataset. In addition, MAE's high mask ratio which forces the model to create meaningful correlations across different patches of an image influenced the joint ViT models. I extend this paper by jointly optimizing classification and reconstruction with the belief that this dual-task approach will enhance feature extraction on small datasets.

CrossViT for Multi-Scale Feature Processing

Chen et al.[2] built on the concept of ViTs with the CrossViT, which addresses the challenge of embedding multiscale feature processing by proposing a dual-branched architecture, with each branch operating on different patch sizes concurrently. One branch operates on fine-grained small patches and the other on larger patches. Each branch is essentially a ViT that independently processes its set of tokens, but there are multiple times where the two branches interact with each other in what is known as cross-attention fusion. These fusion layers allow the large branch to incorporate detailed information from the small branch while also providing the small branch with the global context through token exchange. After several cross-attention exchanges, the outputs of both branches are merged by concatenating the class token embeddings to produce the final classification prediction.

The architecture behind CrossViT's dual branch design is a very innovative way to introduce multiscale feature learning into a transformer. It combines CNN's ability to handle multiscale processing with the transformer framework. CrossViT served as an inspiration by showcasing the value of multiscale processing and feature fusion which prompted me to adapt these principles to address 2 key ViT

limitations: Computation cost and reliance on large datasets, tailoring these trained models on a smaller datasets such as CIFAR-100.

DynamicViT for Dynamic Token Sparsification

Rao et al. [3] introduces the idea of DynamicViTs, which aim to make vision transformers more efficient at inference tasks by dynamically pruning tokens. DynamicViTs introduces small prediction modules at certain layers that estimate an importance score for each token based on its current features. Tokens with low scores are pruned for succeeding layers that can operate on the most informative tokens. They also provide a significant boost in efficiency for ViTs while preserving accuracy. This means that the model runs much faster while achieving a very slight drop in accuracy. Because pruning decisions are learned end-to-end, the model automatically figured out which patches are not needed without any manual intervention. Another strength is that DynamicViTs can be plugged into different ViT architectures without altering their core design, making it a great add-on for efficiency.

DynamicViT is a significant topic as it addresses the issue of efficiency in ViT adoption. My approach builds on token sparsification, integrating learned pruning components to prioritize computation on critical image areas, ultimately paving the way for an efficient and scalable ViT implementation.

Problem Definition

This project addresses a critical limitation of ViTs: **How can ViT architectures be optimized to improve scalability, efficiency, and feature representation for large scale vision tasks, overcoming a very high computational cost and reliance on a large amount of labeled data?** The technical challenges addressed in these papers that arise from this question are as follows: (1) devising different self-supervised learning techniques that can unlock ViTs' feature learning potential without large datasets [1], (2) enabling transformers to capture multi-scale visual patterns for better recognition [2], and (3) reducing the computational cost of ViT by eliminating unnecessary computations [3].

These technical challenges reflect a strong understanding of how to eventually unlock the ability of ViTs to be used in computer vision tasks. The efficiency of the DynamicViT approach can enable greater use in resource-constrained environments. CrossViT's multiscale fusion architecture addresses the complexity of extracting complex patterns from images for better recognition, and MAE's self-supervised approach presents the ability to scale learning to unlabeled datasets, allowing better performance. The research question dives into computational costs and feature extraction as bottlenecks while demonstrating how these solutions present a step in the right direction. By integrating these ideas, the challenge not only defines the objective of the papers but also their relevance in improving the current architecture

of Vision Transformers to meet the demands of modern, large-scale vision tasks with a substantial improvement in accuracy. This integration will create an opportunity for the deployment of ViTs in different real-world applications, such as autonomous driving and medical imaging, where efficiency and rich feature representation are very crucial. These advancements highlight the urgency of developing ViT architectures that can take advantage of self-supervised learning and multiscale processing, addressing the limitations that prevent their widespread adoption.

Methodology

The experimental setup involves training and evaluating four different ViT models: Baseline ViT, MAE pre-trained ViT with fine tuning, Joint ViT with random masking, and Joint ViT with importance masking. The intention is to balance classification accuracy and feature learning through masking and training strategies. I used data loaders to handle shuffling and batching. The batch size was 64 and shuffling was enabled to ensure random sampling and disabled for validation and testing.

Dataset and Model Architectures I used the CIFAR-100 dataset which consists of 50k training and 10k test images. The training set is split into 45k images for training and 5k for validation. During the data preprocessing stage, there are standard augmentation techniques, such as random cropping and normalization, to ensure that the data meets the requirements for training ViTs.

This project implements four ViT based models using PyTorch, all derivations of the small-scale ViT architecture:

- **Baseline ViT:** A standard ViT model which processes images by dividing them into patches which are then embedded into tokens using a convolutional patch embedding layer. A learnable class token is created and augmented with positional embeddings to retain spatial information. The tokens are then processed by the encoder. This model serves as a reference for comparison for the other models.
- **MAE + Fine Tuning ViT:** A ViT model pre-trained with MAE for 50 epochs and then fine-tuned for classification for 50 epochs. It uses the same encoder as the Baseline ViT. The encoder processes these tokens through transformer layers, creating patches which are embedded into tokens. The encoder then processes these tokens. During pre-training, 75% of the patches are masked randomly and a decoder then reconstructs the masked patches. A learnable mask token replaces the masked patches during decoding. After pre-training, the decoder passes the output to the encoder to fine-tune for classification, using the token's output for prediction without masking.
- **Joint ViT with Random Masking:** A ViT model that jointly optimizes classification and reconstruction.

It uses the same ViT encoder and the processing of the tokens follows the same process as the Baseline ViT. During training, 75% of patches are randomly masked which generates random indices to exclude 75% of patches, leaving 25% visible for the encoder. The encoder output simultaneously gets passed to a classification head that predicts class logits and the patch tokens are fed into the decoder that reconstructs masked patches in one training phase. It can be shown as:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \alpha \cdot \mathcal{L}_{\text{recon}} \quad (1)$$

where:

- $\mathcal{L}_{\text{cls}}$: classification loss (cross-entropy)
- $\mathcal{L}_{\text{recon}}$: reconstruction loss (mean squared error)
- $\alpha \in \mathbb{R}_+$: weighting coefficient to balance the two losses

The classification loss is given by the standard cross-entropy formulation:

$$\mathcal{L}_{\text{cls}} = - \sum_{i=1}^C y_i \log(\hat{y}_i) \quad (2)$$

where:

- C : total number of classes
- $y_i \in \{0, 1\}$: one-hot encoded ground-truth vector
- $\hat{y}_i \in [0, 1]$: softmax probability predicted for class i

The reconstruction loss is computed over masked patches:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{M}|} \sum_{j \in \mathcal{M}} \|\hat{\mathbf{x}}_j - \mathbf{x}_j\|_2^2 \quad (3)$$

with:

- $\mathcal{M}$: set of masked patch indices
- $\mathbf{x}_j \in \mathbb{R}^d$: original patch embedding
- $\hat{\mathbf{x}}_j \in \mathbb{R}^d$: reconstructed patch

- **Joint ViT with Importance Masking:** A ViT model similar to the previous joint ViT model but uses importance-based masking. The key difference is the masking logic: the importance masking logic uses a pre-trained ResNet18 model to compute importance scores. Images are passed through ResNet18's convolutional layers and interpolated to an 8x8 grid. Scores are normalized and converted to masking probabilities. The encoder processes the visible patches and the output is gets passed for classification and reconstruction.

Algorithm 1 Joint ViT Training with Random or Importance Masking

Input: Dataset D_{train} , D_{val} , model M , mask_type $\in \{\text{random \& importance}\}$, mask ratio ρ , epochs E , learning rate η , loss weight α
Initialize M randomly; if mask_type = importance, load pre-trained model; move M to GPU.
for $e = 1$ **to** E **do**
 for each batch $(X, Y) \in D_{\text{train}}$ **do**
 Mask ρ -fraction of X :
 if mask_type = random, mask randomly.
 else use importance scores and sampling.
 Compute logits and reconstructed $\hat{X}$ using M .
 Compute L_{cls} (cross-entropy of logits and labels Y), L_{recon} (mean squared error of reconstructed and unmasked patches), and $L = L_{\text{cls}} + \alpha \cdot L_{\text{recon}}$.
 Backpropagate L , update M with AdamW, rate η .
 end for
 Compute accuracy on D_{val} .
 Log L_{cls} , L_{recon} , accuracy.
end for

Experimental Results

This study evaluated four ViT models on the CIFAR-100 dataset which contains 50,000 images and 10,000 test images to compare different training strategies over 50 epochs. Figure 1 shows the results of the classification loss curves between the 4 models over the course of 50 epochs. The Baseline ViT started with a classification loss of 4.04 which decreased to 0.21 by the 50th epoch, while its validation accuracy rose from 9.9% to 41.6%, ultimately resulting in a test accuracy of 41.8%. The MAE pre-trained ViT that was fine-tuned for classification began with a classification loss of 3.92 and ended with a loss of 0.29, along with a validation accuracy of 39.70% and a test accuracy of 43.82%.

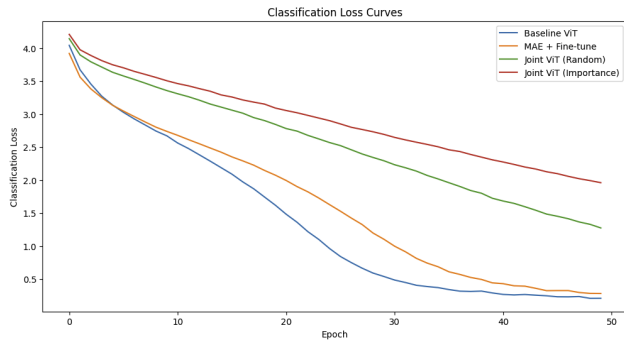


Figure 1. Classification Loss Curve

The Joint ViT models, which simultaneously handled classification and reconstruction tasks in every epoch had higher

classification losses in the early stages of training as shown in Figure 1. The Random and Importance Masking models started with total classification losses of 4.2, decreasing to 1.96 and 1.28 respectively. The joint ViT (Random Masking) model achieved the highest test accuracy of the 4 models with a 43.21% score with a final reconstruction loss of 1.12. The joint ViT (Importance Masking) model came in second place with a test accuracy score of 42.44%, highlighting the superior method of joint training strategies. The general validation accuracy chart during the training of each model is shown in Figure 2.

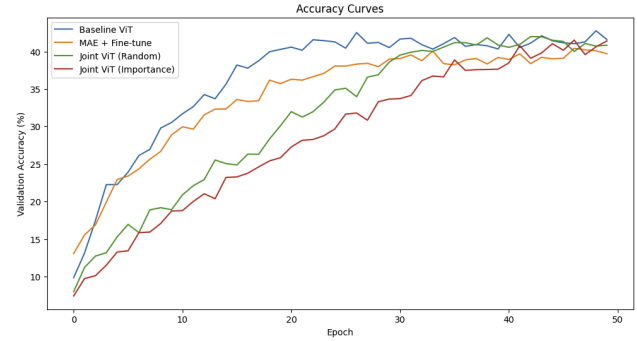


Figure 2. Accuracy Curves

Although each model gradually showed a higher validation accuracy from the first epoch to the final, the epoch-wise validation improvement plot shown in Figure 3 illustrates how the percentage improvement fluctuated between -2% and 5% between each epoch. The variability is evident in all models, although the patterns differ. The AdamW optimizer, used with a learning rate of 0.0005 and a weight decay of 0.05 created a random distribution which led to significant changes in predictions especially during the early epochs, causing the model to overshoot or undershoot optimal weights, resulting in variable improvements. Also during training, the models were updated using mini batches of 64 images, introducing noise as each batch could have emphasized different features, also causing fluctuations in accuracy improvements.

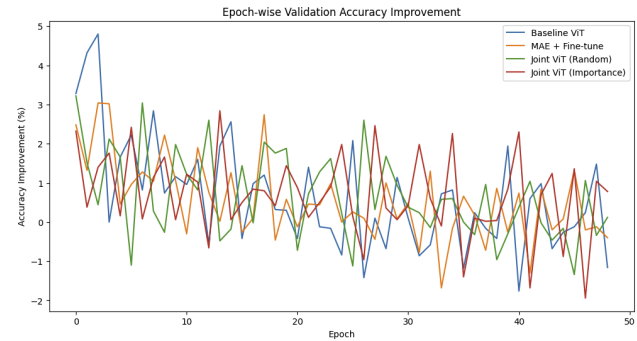


Figure 3. Epoch-wise Validation Accuracy

The training dynamics across each of the models was further explored through the relationships between validation accuracy and classification loss as depicted in Figure 4. This scatter plot illustrates how the 4 models displayed their highest accuracy at their lowest classification loss. The spread of the joint ViT models points across a wider range of losses and accuracies suggest larger variability due to the objectives of classification and reconstruction. This highlights the tradeoff in joint training where higher losses are tolerated for better generalization. This contrasts with the Baseline ViT and MAE + fine tuned model which have a much tighter cluster, indicating more stability but less adaptive learning during the training process.

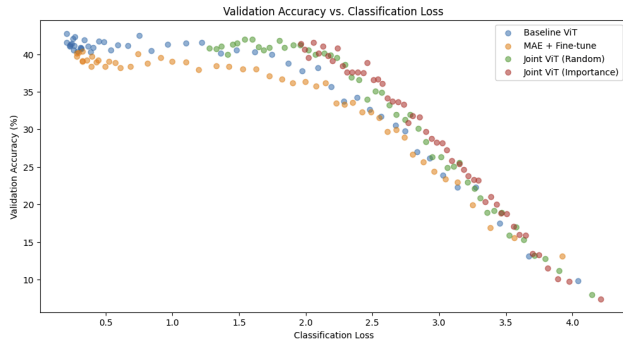


Figure 4. Validation Accuracy vs Classification Loss

Contributions

In this project, I extend the work discussed in He et al. in their paper, "Masked Autoencoders are Scalable Vision Learners" to explore different training strategies for ViTs on the CIFAR-100 dataset. My work builds upon the MAE concept of self-supervised learning where a model is trained to reconstruct patches of an image that are randomly masked, but I expand on this idea using a small dataset.

My primary contribution is the implementation of a joint training approach that optimizes classification and reconstruction within each epoch, implemented in both the joint ViT random masking and importance masking models. Unlike the original approach, which separates pretraining (reconstruction) and fine-tuning (classification) into distinct stages, my approach combines these two objectives. The joint training strategy allows the ViT to learn features for classification tasks while reconstructing masked patches concurrently, which drastically reduces the computation overhead compared to the MAE + fine-tuning approach.

My second contribution is the introduction of importance masking in the joint ViT (Importance) model, which incorporates an importance-based masking strategy. Instead of randomly masking patches on an image, I use a ResNet18 model to calculate importance scores for each patch, masking 75% of the images based on these scores. This masking technique introduces a dynamic selection process between

different image regions that could be enhanced in future iterations

Conclusion and Future Directions

This project explored four different ViT architectures for image classification on the CIFAR-100 dataset with the goal of addressing the challenges of high computational cost and reliance on large datasets. The models evaluated were: a Baseline ViT trained for classification tasks, a MAE + fine tuned model pretrained for 50 epochs and fine-tuned for classification for 50 epochs, a joint ViT (Random) model that simultaneously optimizes classification and reconstruction of random patches in an image, and a joint ViT (Importance) model that expands on joint training but uses an importance-based masking algorithm utilizing an initialized ResNet18 model.

The joint ViT Random model achieved the best classification accuracy among the four models in this study, supporting the belief that simultaneously handling classification and reconstruction can improve the overall generalization. The joint ViT importance model shows really strong potential as well as the final accuracy was very close to the random model. A key takeaway from this project is that ViTs can benefit greatly from self-supervised learning even when applied to a small dataset like CIFAR-100. The full potential of joint ViT models has not been realized yet. This study showed slow convergence and high variance in the loss curves which could be improved with a different training technique.

There are several directions to extend this research from here. While this research was constrained to 50 epochs on a Tesla T4 GPU, more extensive pre-training or using larger ViT models could help push the scalability and further unlock the joint training strategies. There could also be further improvements expanding on the end-to-end learning of the importance scores by implementing the masking logic directly into the ViT architecture instead of using a pretrained model for importance masking and randomly masking 75% of the patches for the random model.

References

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